1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L14 – Prototyping**

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3 **Recap**

- Earlier we learned requirement gathering through CI and design with guidelines
- This lecture we shall learn prototyping techniques
- First let us recall SDLC for UCD

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6 **Prototyping for Feedback**

- The guidelines offer merely a starting point - once we have those, we can go for a design

7 **Prototyping for Feedback**

- However, to know if the design is going to work in practice, we need not fully implement the idea first

8 **Prototyping for Feedback**

- Instead, we go for a prototype, a lightweight implementation in a sense, to test our idea

9 **Prototyping for Feedback**

- In UCD, prototypes play very important role
- Built typically at a very early stage of the development life cycle

10 **Prototyping for Feedback**

- Once its purpose is served, a prototype can be discarded

11 **Prototyping for Feedback**

- In the best case, the prototype can be incrementally refined (and tested) as we make progress in the development till the end, at which point it becomes the fully-implemented product in itself

12 **Prototype Categories**

- Horizontal - the entire interface is depicted at the surface level without any

functionality
- Interaction is not prototyped
- No real work can be done
- Suitable to discuss, brain-storm or elicit feedback on the interface look and feel, primarily

13 **Prototype Categories**
- Vertical - designed to represent interaction
  - Few selected features implemented in-depth, starting from the first screen to the screen after the last action is performed
  - Suitable for analysis of interactions and features

14 **How to Create Prototypes**
- Low fidelity
  - No technological intervention - made with paper, cardboards, woods and so on
  - A popular example are the paper mock-ups for interface look, feel and even functionality
  - Quick and cheap to make and modify
  - Good for horizontal prototypes and used to brain-storm on alternative designs and get user response on the idea

15 **Low Fidelity Prototype: Interface Sketches**
- Interface sketches offer another low-fidelity prototypes
  - Drawings depicting major components of an interface
  - Provide way to envision appearance of the interface

16 **Low Fidelity Prototype: Interface Sketches**

Example - consider a shop that sells various consumer items (clothes, bags and so on). They want an interactive system to generate bills based on the selection by a customer. The customer selection is "scanned" and identified by the system and a bill is generated. A sketch for the proposed interface design is shown along with one possible final design.

17 **Low Fidelity Prototype: Interface Sketches**

18 **Low Fidelity Prototype: Interface Sketches**

19 **Low Fidelity Prototype: Interface Sketches**
- Sketches in itself are not sufficient to prototype interaction
  - From the previous sketch, it is not possible to know the sequence of actions and the corresponding changes on the interface to execute a task (a "stroller purchase" in this case)

20 **Low Fidelity Prototype: Interface Sketches**
- Therefore, sketches essentially represent horizontal prototypes

21 **Low Fidelity Prototype: Interface Sketches**

- What we need is not a single sketch, but a series of sketches, often called "storyboarding" to represent interaction

22 **Low Fidelity Prototype: Storyboarding**

23 **Low Fidelity Prototype: Storyboarding**

24 **Low Fidelity Prototype: Storyboarding**

25 **Low Fidelity Prototype: Storyboarding**

26 **Low Fidelity Prototype: Storyboarding**

27 **How to Create Prototypes**

- Medium-fidelity - when we use computer to create simple (low fidelity) prototypes
  - If the computer is used to create static sketches only, it still remains horizontal prototype since no real functionality is there

28 **How to Create Prototypes**

- Medium-fidelity prototypes may include videos as well

29 **How to Create Prototypes**

- Ex - we can have an animation video (created with Adobe Flash tool, for example) to prototype the food selection app of the previous slide
  - In the video, simple controls can be provided to simulate the interaction

30 **Medium-Fidelity Prototyping**

- In fact, the interaction can also be simulated with as simple a tool as a Microsoft PowerPoinT™ slideshow
  - The key frames can be converted to slides
  - With simple controls (e.g., timers or key press), slide transition takes place depicting the interaction
  - In that way, the storyboard can be converted to a vertical prototype

31 **How to Create Prototypes**

- Hi-Fidelity Prototypes – prototypes created with computer programs (e.g., actual software development)
- More sophisticated and requires much more effort (including expertise in programming) than the other two categories

32 **How to Create Prototypes**

- There are many interface builders, "toolkits" and "wizards" to ease the programming efforts

33 **How to Create Prototypes**

- Popular examples include the Tcl/Tk toolkit, Visual Basic programming language, JAVA Swing library and so on

34 **How to Create Prototypes**

- The toolkits and libraries provide support for "widgets" (GUI elements) - a programmer can directly use those (rather than creating on his/her own) and build GUIs

35 **How to Create Prototypes**

- The interaction is implemented through programming

36 **How to Create Prototypes**

- Hi-fidelity prototypes are mainly used to create vertical prototypes

37 **How to Create Prototypes**

- "Wizard of Oz" approach - an interesting prototyping technique

38 **How to Create Prototypes**

- In this technique, a human subject *believes to* interact with an *autonomous* computer
- In reality, however, the computer is operated by an "unseen" human being (the "Wizard")
- The term originated in the early 1980s, although the concept was there beforehand

39 **How to Create Prototypes**

- Example - the "listening typewriter" experiment conducted by the IBM in 1984
  - IBM was trying to develop a speech recognition system
  - They wanted to test the acceptability of the idea, requiring a working system
  - Problem - nobody had the speech recognizer and the company did not want to spend money and effort before testing the acceptability

40 **How to Create Prototypes**

- So, they decided to go for a "Wizard of Oz" experiment

41 **Wizard of Oz Experiment**

42 **How to Use Prototypes**

- It can be "thrown away," i.e., discarded after use (Cardboard mock-ups and paper sketches suitable for such treatments, since these are low-cost)
  - We should be able to create such prototypes rapidly (otherwise, it becomes too expensive to discard a prototype made over a period of time with considerable man-hour spent)

43 **How to Use Prototypes**

- In the "incremental" approach, the system is designed into units (modules)
  - Each unit is separately prototyped and tested
  - Afterwards, it is integrated into the system

44 **How to Use Prototypes**

- The other approach is "evolutionary" – the whole system is prototyped and tested
  - Based on the testing, the prototype is altered
  - Eventually, it becomes the final product

45 **Note**

- We discussed few of the SDLC stages (requirement gathering, design with guidelines, prototyping)

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47 **Note**

- Other important stage is early evaluation – we shall discuss next

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49 **Book**

- Bhattacharya, S. (2019). Human-Computer Interaction: User-Centric Computing for Design, McGraw-Hill India
  - Print Edition: ISBN-13: 978-93-5316-804-9; ISBN-10: 93-5316-804-X
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